THE UNITED REPUBLIC OF TANZANIA

ORGANIZATION OF HEADS OF NON-GOVERNMENT SECONDARY SCHOOLS

(OHONGSS)

KIGOMA REGION

FORM FOUR MOCK EXAMINATION

031/1 **PHYSICS 1**

TIME: 3 HOURS **YEAR: 2024**

Instructions

1. This paper consists of sections A, B and C with a total of **eleven** (**11**) questions.
2. Answer **all** questions in sections A and B and only **two** (**2**) questions from section C.
3. Cellular phones are not allowed in the examination room.
4. Non programmable and scientific calculators may be used.
5. Where necessary, the following constants may be used.
   (i) Acceleration due to gravity on earth, $g_e = 10m/s^2$.
   (ii) Specific heat capacity of water = 4200 J//Kg K.
   (iii) Speed of light in air = 3×10^8 m/s.
   (iv) Speed of sound in air = 340 m/s.
   (v) Speed of sound in water=1500m/s
   (vi) Coefficient of linear expansion of aluminium = $2.6x10^{-5}/^0C$
   (vii) Specific latent heat of vaporization of water = $2.268x10^6$ J/kg
   (viii) Atmospheric pressure=101Kpa
   (ix) Linear expansivity of brass=$19x10^{-6}/K$
   (x) Linear expansivity of iron=$12x10^{-6}/K$

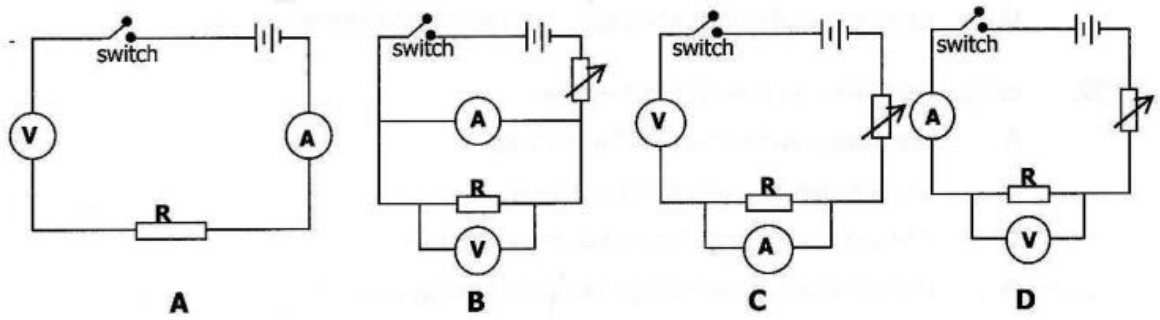
MsomiBora.com

SECTION A (16 Marks)

Answer all questions in this section

1. For each of the items (i) – (x), choose the correct answer among the given alternatives and write its letter beside the item number.

(i). Which circuit diagram can be used to verify ohm's law?



(ii) Diffusion occurs more quickly in a gas than in a liquid because;-

A. The liquid contains a layer on its surface
B. The gas contain semi-permeable membrane
C. The gas molecules is small in size compared to the liquid molecules
D. The adhesion is large than cohesion in gas compared to that in liquid.
E. The speed of molecules in gas is greater than in liquid.

(iii) In a loading a lorry a man lifts boxes each of weight 100N through a height of 1.5m, if he lifts 4boxes per minute, the average power the man is working is;-

A. 100w B. 10w C. 600w D. 37.5w E. 2250w

(iv) A physics student stands some distance from a cliff. She claps her hands and half a second later hears an echo. Taking the speed of sound in air as 320m/s, how far from the cliff is she standing?

A. 80m B. 160m C. 320m D. 670m E. 870m

(v) A 24Ω resistor is to be connected in series with a 12 V battery. What is the power loss in the resistor?

A. 0.5W B. 6W C. 12W D. 24W E. 30W

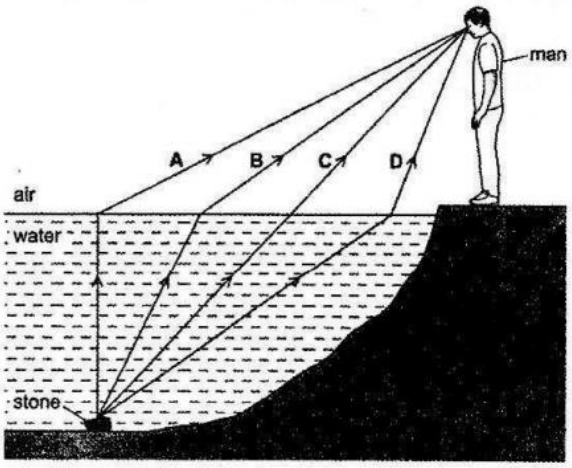
(vi) The acceleration of a moving object may be found from

A. The area under its velocity-time graph.
B. The slope of the velocity-time graph.
C. The area under its distance-time graph.
D. The slope of the distance-time graph.
E. The slope of the peak of its distance-time graph.

(vii) Which of these resources of energy is non-renewable?

A. Wave energy B. Bio fuels C. Radiant energy D. Fossil fuel
E. Geothermal energy

(viii) A man sees a stone at the bottom of a pool of water



Which path, A, B, C or D could be taken by light from the stone to the man?

(ix) Light waves differ from sound waves because;

A. Light is an electromagnetic wave
B. Light wave are long and sound waves are short
C. Interference is obtained with light waves but not with sound waves.
D. The speed of light is independent of the medium in which it travels
E. Sound waves do not travel in water

(xi) A certain substance is radioactive. If $15/16^{th}$ of the radioactive atoms of the substance decay in 8minutes,determine the half- life of the substance

A 32min B 16min C 4min D 2min E 1min

2. Match the items in **List A** with those in list B and write a letter of the item from **List B** beside item number in list A.

LIST A	**LIST B**
(i) Presbyopia	A. Close objects appear clearly, but far ones do not
(ii)Hyperopia	B. Imperfection in the curvature of the eye that causes the formation of blurred images
(iii)Myopia	
(iv) Astigmatism	C. Gradual loss of the eye's ability to focus on nearby objects, due to old age
(v)CCTV cameras	D. Images of the nearby objects are blurry
(vi) Accommodation	E. Allow us to see objects that are far away
	F. Refraction or reflection of light to form different types of images
	G. Eye focuses on objects at different distances by varying the focal length of the lens
	H. Surveillance in high security installations, banks, shopping malls, offices and residences

SECTION B (54 Marks)

Answer **all** questions in this section.

3. (a)(i) What causes the build up of dust on a television screen?.(2marks)

(ii) Explain how fish can survive under water when the surface is already frozen (2 marks)

(b) (i) A man holds a newspaper at arm's length to read it, use diagrams to show the defect of his vision and how he should correct it (2Marks)

(ii) Peter stands at a distance of 5.0m from the hole of a pinhole camera. If the distance of the film from the hole is 20cm and an image 5.0cm long is obtained on it, find the height of peter. (3 marks)

4. (a)(i) Explain why ice skates have narrow blades.(2 marks)

(ii) Why do humans have two eyes? (2 marks)

(b) (i) when a ship sails from fresh water into the sea, does it sink deeper into the water or rise a little out of it? Explain your answer (2Marks)

(ii) A uniform metre rule of uniform width 2.5cm and thickness 0.5cm is suspended at the 80cm mark and kept balanced horizontally by hanging a mass of 150g at the 100cm mark, calculate the density of the material of the metre rule and the tension, T, in the string. (3marks)

5. (a) (i) When the bulb of the thermometer is dipped in hot water, there is a fall in the mercury level at the beginning and then it begins to rise. Explain (2 marks)

(ii) Explain why cups of soap at a take-away cafe are often sold in white, polystyrene cups with a lid to stop spillage.(2 marks)

(b) (i) A hollow metal sphere floats with 7/8 of its volume submerged in water of density 1000kg/m3 and 5/8 of its volume submerged when transferred to another liquid, calculate the density of the liquid(2 marks)

(ii) An athlete runs at 4m/s from point A to point B, immediately turns and runs back from B to A with a speed of 8m/s. calculate the average speed and velocity of the athlete. (3 marks)

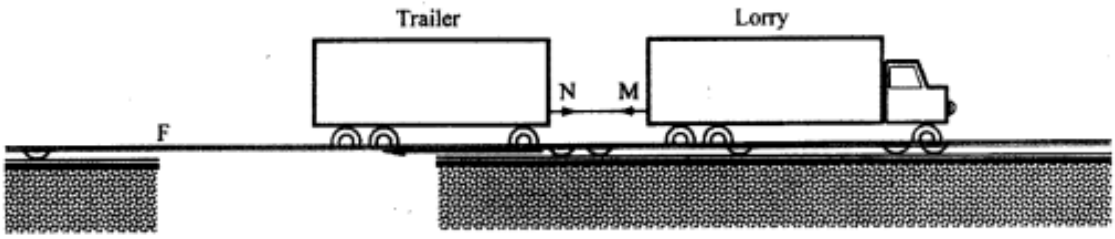
6. (a)(i) Explain why the wet clothes put in a drum get dried faster when the drum of the drying machine is rotated at high speed (1marks)

(ii) A human being can exert surprisingly large amounts of power but for short intervals. You are asked to determine your power output in running up a flight of stairs, state what measurements you need to know and how you would measure them (1.5 marks)

(b) (i) A room lamp is connected to a switch, it is desired to insert a second lamp so that both lamps light normally when the switch is closed .Draw a circuit diagram to show how this could be done(1.5 marks)

(ii) The diameter of an aluminium sphere at 200C is 2.5cm. If its diameter increases to 2.51cm when heated, what is the final temperature? (3 marks)

(iii) The figure below shows a lorry towing a trailer using rope



The lorry exerts a force N on the trailer and the trailer exerts an equal but opposite force M on the lorry. The frictional force between the trailer and the road is F. Explain how the force N, M and F enable the trailer to move. (**2marks**)

7. (a) (i) Professor Mfakunoga prepared tea for her friends ,unfortunately they didn't go early, what device would you advise her to buy so that she maintains the hotness of the tea? With the aid of a diagram, explain the features of that device which enhances its efficiency. (**3 marks**)

(ii) Explain why people are advised against exposing themselves to x-rays unless it is absolutely unavoidable. (1.5 marks)

(b) (i) An induction coil is in fact, a step-up transformer designed to be used with a d.c .Explain how the changing magnetic field is produced using a d.c in the primary coil (1.5marks)

(ii) A 2.0 KW immersion heater is used to heat 4.0x10-3m3 of water from 200 C to 1000C . Given that 30% of the heat is lost to the surroundings, and the density of water is 1000Kg/m3, calculate the time needed for the heating process (3 marks)

8. (a) (i) A sealed plastic bottle filled with water to the brim breaks after some time if it is placed in the freezing compartment of a refrigerator. Explain (1.5 marks)

(ii) In spite of the disadvantages of nuclear power stations, most people agree that nuclear power is worthwhile. Explain why (1.5 marks)

(b)(i) A vibration string is sounded with 288Hz tuning fork and 3 beats are counted per second. The string is then loaded with small amount of wax and 2 beats are counted per second. What is the frequency of the loaded string? (3marks)

(ii) The effective resistance of 2 resistors connected in series is 25Ω. When connected in parallel, the effective resistance of the combination is 6Ω. Calculate the values of the 2 resistors (3 marks)

SECTION C (30 Marks)

Answer only **two** (**2**) questions in this section.

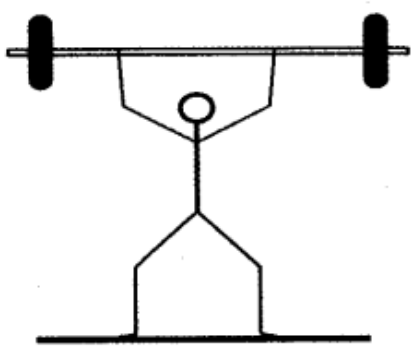
9. (a) (i)The fundamental frequency of vibration of a string is **f**. What will the fundamental frequency of the string be if the length of the string is halved and the tension is increased four times? (3 marks)

(ii) Explain why transmission cables carrying high voltages are supported high above the earth by pylons (2 marks)

(b) (i) Explain why α-rays source is more dangerous when inside a human body than outside? (2 marks)

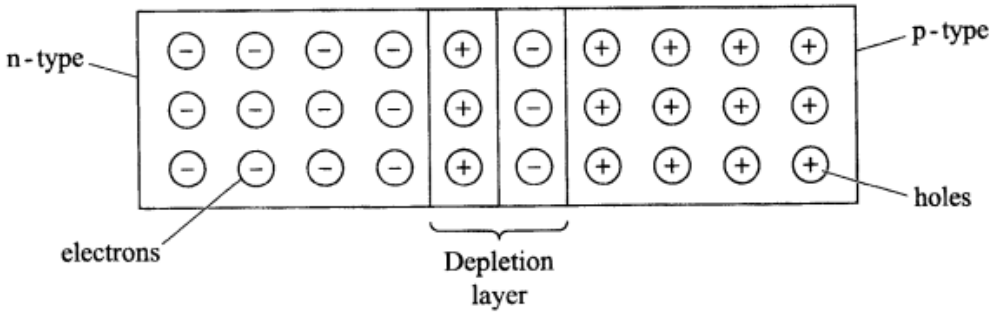
(ii) A fresh sample of a tree gives 240 counts/minute and sample under test gives 15 counts/minute. Given that the half-life of carbon-14 is 5500 years, estimate the age of the sample. (3marks)

(c). A Figure below shows an athlete lifting weights while standing with the feet apart.



Explain why standing with the feet apart improves the athlete's stability. (5 Marks)

10. (a) Figure below shows a depletion layer in an unbiased p-n junction.



State how a battery can be used to make the depletion layer narrower (**4 marks**)

(b) Draw and label a diagram of a GEIGER MULLER TUBE (GMT). Explain how it can be used to determine the background radiation at your school (6 marks)

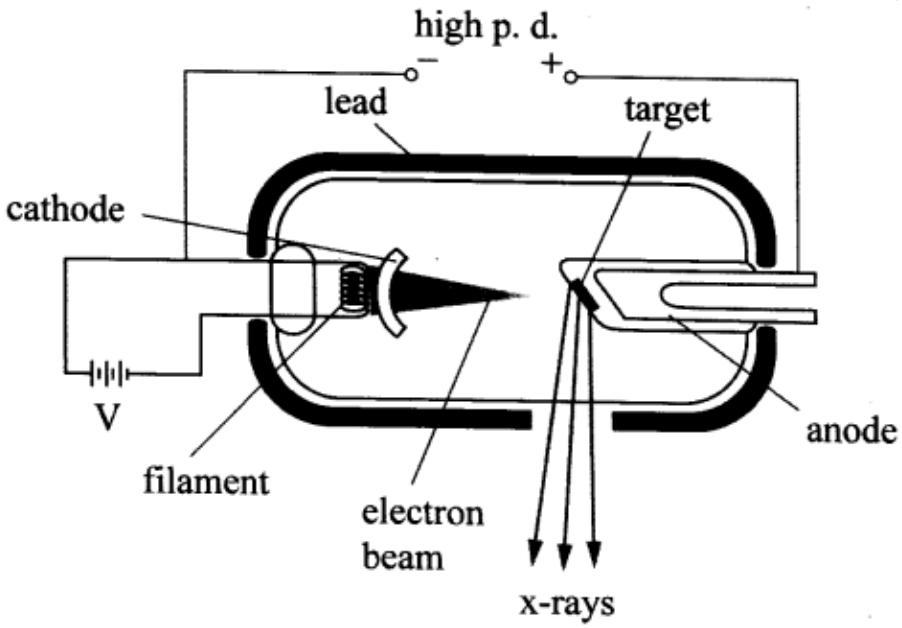
(c) The difference in length between a brass and an iron is 14cm at 10^0C. What must be the length of the iron for this difference to remain at 14cm when both rods are heated to 100^0C? (5marks)

11. (a) (i) A ship using an echo-sounding device receives an echo from a wreck 0.8 second after the sound is transmitted. If the velocity of sound in sea-water is 1500m/s, determine the depth of wreck. (3 marks)

(ii) Why can a microwave oven cook much more quickly than a conventional one? (2 marks)

(b) A rate meter records a background rate of 7 counts per second. When a radioactive source is held near, the count rate is 167 c/s. If the half-life of the source is 7 days, what will the recorded count rate be 21 days later? (5 marks)

(c) Figure below shows parts of an x – ray tube.



Explain why:

(i) A potential difference is applied to the filament (2 Marks)

(ii) A high potential difference is applied between the cathode and anode (1.5 Marks)

(iii) Most of the tube is surrounded by lead. (1.5 Marks)